

1. Surface Discovery

BFS Web Crawler

- Link & Form Extractor
- Query String Parsing
- Depth Bounded (D_{max})

OpenAPI / Swagger

- v2.0 / v3.0 Spec Parser
- Schema & Path Variables
- HTTP Verb Registration

2. Scanner Engine (14 Plugins)

ThreadPoolExecutor ($k = 8$) | Baseline Differential Cache

Reconnaissance

Nmap, Nikto, DirBrute

Web Injections

SQLi, XSS, CORS, Misconfig

API & Auth

APIAbuse, Auth, JWT

Exploit Vectors

CmdInj, SSRF, PathTrav, SSTI

3. Intelligence & Correlation Layer

Heuristic Deduplication

Compound key: host+port+URI+weakness

Threat Enrichment

Live queries to NIST NVD & CIRCL APIs

Taxonomy Mapping

Deterministic OWASP, CWE & PCI-DSS tags

4. Delivery, CI/CD Policy Gating & Multi-Channel Reporting

CI/CD Build Gating

Exit 1 on threshold ($\geq$ threshold)
Non-zero exit on criticals

Set-Theoretic Delta

Partitions Delta_NEW, Delta_FIXED
Tracks regressions vs baseline

Multi-Channel Output

Real-Time SSE Web Dashboard
PDF / HTML Reports & Webhooks